Godofredo Alves

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Hands-on Principal Full-Stack Architect / Technical Lead with 20+ years designing, building, and delivering production-grade applications with Java 17/Spring Boot, Python, React, and AWS. Currently providing technical leadership across 3 teams (17 engineers) building an AI-powered financial/portfolio intelligence SaaS: microservices, REST APIs, credit and risk data, and LLM integration. Strong AI-enabled engineering practitioner with a financial services background (Santander, Mapfre) and deep expertise in code quality, testing, CI/CD, and architecture.

Work Experience

AtrioIQ is an AI-powered Portfolio Intelligence SaaS platform that gives self-directed investors institutional-grade, explainable research filtered through their own holdings.

Principal Architect / Full-Stack Tech Lead at AtrioIQ **Oct 2025-Present**

- Hands-on Principal Architect and technical lead for 3 engineering teams (17 engineers: Platform & Backend, AI & Data, Frontend & QA).
- Lead the design and development of an AI-powered financial/portfolio intelligence SaaS built on Java 17/Spring Boot, Python, React, and AWS.

Key Achievements: Designed and delivered the platform architecture end to end, including the AI "Senior Analyst" engine and the Java 17/Spring Boot credit-risk service, and increased delivery speed through AI-enabled development practices.

Activities descriptions:

- Architected a microservices platform: React/TypeScript SPA, Node.js/Express API gateway (auth, routing), Python FastAPI portfolio & analytics service, and Java 17/Spring Boot financial data and risk services, communicating via REST and event-driven messaging (SQS/SNS).
- Designed and built the Java 17/Spring Boot Financial Data & Credit Risk service on AWS (ECS Fargate, SQS, RDS PostgreSQL, ElastiCache): ingests and normalizes SEC EDGAR/XBRL filings and a 5-provider market-data ladder (AlphaVantage, Finnhub, Yahoo, etc.), computes credit and risk metrics (leverage, interest coverage, debt/EBITDA, volatility, drawdown, Sharpe, concentration exposure), and publishes them to the AI engine.
- Designed the "Senior Analyst" AI engine: multi-provider LLM orchestration (Claude, OpenAI, Gemini) with prompt engineering, structured JSON output validation, retrieval-augmented generation (RAG) over an investment-methodology knowledge base, and evaluation caching.
- Java service engineering: Spring Data JPA, Spring Batch for scheduled filing ingestion, Resilience4j circuit breakers and rate limiting for third-party APIs, data-quality validation with full lineage and audit trail, JUnit 5/Testcontainers, and OpenAPI contracts.

- Built explainable, portfolio-aware AI features: company deep analysis, news and sentiment impact, technical chart-setup scanner, thesis tracker, and AI-generated portfolio reports (Excel/Word).
- Enforced AI guardrails: no hallucinated values (missing data is labeled), no buy/sell instructions, mandatory disclaimers, and demo-mode isolation of synthetic data.
- Introduced AI-enabled development across the teams (Claude Code, Codex, AI agents with shared project memory, plan/task governance), noticeably increasing delivery speed while keeping quality gates.
- Built CI/CD with GitHub Actions, Docker, and Cloud Run/AWS deployments; quality gates with pytest, Vitest, Playwright E2E, linting (Ruff), pre-commit hooks, and performance probes.
- Data layer with PostgreSQL (Alembic migrations) and SQLite for local development; JWT authentication and multi-user SaaS readiness.
- Mentor engineers and tech leads through design reviews, pair programming, and code reviews; set coding standards and the definition of done.
- Drive architectural decisions with architecture decision records (ADRs) and translate product and business requirements into practical technical designs.
- Troubleshoot complex production issues and lead performance, scalability, and reliability improvements.
- Technologies: Java 17, Spring Boot, Python, FastAPI, TypeScript, React, Node.js, PostgreSQL, Docker, AWS, GCP, GitHub Actions, LLM APIs.

Synopsys Inc.

Black Duck

Synopsys Inc. is a leading provider of electronic design automation solutions and services. After a spinout in 2024, the company became Black Duck and was acquired by Clearlake Capital Group and Francisco Partners

Software Architect / Senior Staff Engineer at Synopsys Inc./ Black Duck 2021-Oct 2025

Closely contributed to improving Black Duck's audit process and internal tools, and mentored JRs.

- Perform Software Quality, Design, and Process Audits, which include the following:
- Review of customer development procedures, including manuals and technical documents.
- Code quality and architectural analysis. Review and report on code metrics to evaluate code quality, improve agility, and reduce bugs, costs, and risks.
- Develop a detailed understanding of the capabilities and limitations of the customer's development architecture and tools.
- Deep insights into a software system's design or architectural health, the likely cost of ownership, and the resulting risk-related consequences.
- The audit offers a complete analysis of the processes and practices that compose the software development life cycle (SDLC). Conducting in-depth interviews with a small number of key personnel to gain insight into the quality and maturity of development practices, including coding standards, processes, and tools. From this, it provides recommendations for improving code quality while reducing development and maintenance costs.
- Overall architecture in terms of modularity and hierarchy, thus rounding out a complete picture of the health of the software. The report includes an analysis of how the architecture impacts maintainability and identifies potential risk areas that are candidates for code refactoring.
- Make recommendations to improve the process, code, design, architecture, etc., before selling, merging, and acquiring.

- Contribute to the improvement of the Audit process (create new tools, automation, etc.).

Agua Limpa is a chemical firm for laboratory and analysis.

Architect Lead / Senior Software Engineer at Agua Limpa. 2018-2021

- Solution Architect and Full-Stack developer with leading experience. Helping customers find their way to the best solution and deliver amazing experiences and software.
- Implemented systemic changes to leverage technology to support business needs effectively, and applied technical design and programming skills.

Key Achievements: I worked on coding, design, and architecture from the beginning of the application, collaborating with the team and stakeholders.

Activities descriptions:

- Day-to-day Follow-up meetings are held for project discussion and follow-up via Jira and Confluence.
- New feature development, existing feature enhancements, and bug fixes.
- Merge control. Peer code review and assistance.
- Software Developer in virtualization platforms, server virtualization, and cloud services.
- Proficient understanding of code versioning tools, such as Git.
- Strong understanding of MVC, other design patterns, and object-oriented design. Programming, and understanding fundamental design principles behind a scalable application.
- Strong understanding of front-end technologies, such as JavaScript, HTML5, CSS3, and Bootstrap.
- Frameworks: JavaScript as Angular, React, Vue.js for the front-end.
- Using Node/Java/PHP as the back end Node/Java/PHP (Servlets, EJB, Spring Boot, Jenkins, Docker, Windows, Linux, AWS, Hibernate connecting with MySQL.
- Creating database schemas that represent and support business processes.
- Strong understanding of SQL/NoSQL databases and their declarative query languages.
- Test-Driven Development (TDD), data mocking, and unit/integration test automation.
- Infrastructure architecture, enterprise network, and microservices architecture.
- Integration of multiple data sources and databases into one system.

Everis is a consulting firm for Business and Information Technology

Senior Software Engineer at Everis/NTT Data Consulting 2015-2018

- Conducted migration and enhancement of data processes for system solution change.
- Designed innovative solutions to improve performance and lower cost in software and prototype production. Offered different software alternatives.
- Guided teams of developers on transitioning from traditional project delivery to a product-based approach; connecting software delivery activities to measurable business results.

Key Achievements: Ability to learn fast and adapt to technology. Focus on delivering high-quality software and care deeply about code quality, readability, and reliability.

Activity descriptions:

Able to work on projects in different companies, since Everis is a consulting company.

- Mapfre S.A. is a Spanish multinational insurance company serving nearly 30 million customers. For this project, I worked as a Solutions Architect/Lead, leading a team of 5 people with different skill sets, from junior to senior. Building a cognitive chatbot that communicates with IBM Watson. The API was intended to automate server, database, backup, stop, and restart processes through application groups like WhatsApp and Telegram. Using Node/Java Spring Boot as the backend, Hibernate connecting with MySQL, Jenkins, Docker, and AWS.

- Santander Bank operates across Europe, South America, North America, and Asia. For this project, I worked as a Solutions Architect/Lead, leading a team of 5 people with different skills from junior to senior. DevOps and automation of environments to support development, testing, and delivery of new software versions. Using JavaEE, .NET, C#, Python, Node, Jenkins, Docker, AWS, Lua, Puppet, PowerShell, Kibana, Hadoop, Git, Git Lab.

- Telefónica Telecommunications is a Latin American company serving nearly 100 million customers. For this project, I was an Architect/developer and leader, leading a team of 7 people with different skills, from junior to senior. Goal: System integration to connect to the Amdocs legacy billing and other systems. Using Java Spring with microservices, Java complements the Oracle Service Bus. As well as BPEL, SOA architecture, Governance, and Web services orchestration.

GLR/POPULIS is a firm for Technology and Human Resources

System Developer at GLR/POPULIS System Consulting 2008-2015

- Full-stack developer based on JavaEE, EJB, HTML, Struts, JSP, Servlet, WEB Services, JS, .NET, and PHP, in a multilayer MVC model using SQL, Oracle, and MySQL databases.
- Participated in the requirements of business analysis, technical analysis, and working with stakeholders.
- Full Java stack: Developing, maintaining, and integrating code for various client software systems and applications
- Performed business analysis to define system requirements.
- Developed and deployed technologies.
- Developed new software functionalities. Created improvements and fixed bugs.

Key Achievements: Successfully developed using Java and PHP. Developing, maintaining, and integrating code for various client software systems and applications.

Technical Skills

In-depth, hands-on experience with:

AI & LLM:
- Claude, OpenAI, Gemini APIs
- Prompt engineering
- RAG / structured output

Servers:
- Liberty
- Apache Tomcat
- Glassfish

Software Development:

Back-end
- Java 17 / Spring Boot
- PHP

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- AI agents (Claude Code, Codex)

DevOps:
- Puppet
- PowerShell
- Docker
- Hadoop
- Jenkins
- Git, GitHub Actions
- AWS (ECS, SQS, RDS), GCP
- Chef
- Lua
- VMware

Management & Documentation:
- Jira
- Trello
- Visual Studio
- Microsoft Project
- Swagger

Data Base:
- MSSQL
- MySQL
- Oracle
- NoSQL
- Elasticsearch

SOA:
- SOA architecture
- ESB bus
- SOA Suite 12C
- WebMethods
- Oracle Service Bus (OSB)
- BPEL
- Data Governance
- Canon Model
- TAM Governance Pad
- Web Services (SOAP, RESTful)
- JBoss
- WebSphere

Authentication:
- JWT, GWT, SSO

Front-end
- Angular, React, TypeScript
- Webpack, Protractor
- Karma
- Django
- HTML5, CSS, Bootstrap
- AJAX
- XML

Back-end
- Python (FastAPI)
- WSDL, XPath, XQuery, Soap, Rest
- Microservices
- .Net
- C#

Framework
- Struts, JSF, Spring
- Wordpress
- Magento
- JavaScript, jQuery
- JSP

Skills

- Strong analytical and logical concepts supported by extensive experience in different technologies, enabling me to quickly adapt.
- Proficient in communication with clients and stakeholders at all business levels.
- Effective facilitator, capable of building consensus, establishing rapport, influencing, and motivating people towards high performance.
- Experience with using new technologies and improving processes to reduce costs.

Education

Post-Graduate Certificate in Project Management
Centennial College – 2019

Bachelor's Degree in Information Systems
University Bandeirantes – 2010